

SatQuery AI — Product Requirement Document

Geospatial Vision-Language Intelligence Platform

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1. Executive Summary & Vision

SatQuery AI is a next-generation geospatial vision-language intelligence platform designed to extract actionable insights from satellite and high-resolution aerial imagery. Powered by a fine-tuned vision-language model (**Qwen2.5-VL-3B-Instruct**) combined with a domain-specific LoRA adapter (**satquery-ai-vqa-lora**), SatQuery AI transforms complex satellite imagery into natural-language answers, detailed descriptions, and localized object bounding boxes.

The primary goal of SatQuery AI is to eliminate manual visual inspection overhead for defense analysts, urban planners, environmental researchers, and disaster recovery teams by enabling conversational visual query processing with sub-3-second inference latencies and zero-cloud dependency options.

2. Strategic Objectives & Success Metrics

Objective	Target Metric	Validation Method
High VQA Accuracy	Top-1 Accuracy > 88% on remote sensing benchmarks	Automated benchmark evaluation on RSVQA & AID datasets
Low-Latency Inference	< 2.5s inference time on standard GPU (4-bit NF4)	Uvicorn response timer logging & /api/health telemetry
Low Hardware Footprint	< 6.0 GB VRAM requirement for execution	bitsandbytes 4-bit NF4 quantization validation
Realtime Synchronization	Sub-500ms sync delay over Firebase Realtime DB	End-to-end event stream benchmark

3. Functional Feature Specifications

Feature 1: Visual Question Answering (VQA)

- **Capability:** Accepts a satellite/aerial image (base64 or multipart upload) alongside a natural language question (e.g., 'How many industrial tanks are located in the north sector?').
- **Endpoints:** POST /api/vqa, POST /api/vqa/upload
- **Output:** Concise, context-aware natural language answer + confidence score + processing timing.

Feature 2: Automated Satellite Image Captioning

- **Capability:** Generates comprehensive textual descriptions covering terrain type, land usage, building density, water bodies, and vegetation index.
- **Endpoints:** POST /api/caption, POST /api/caption/upload
- **Output:** Detailed paragraph describing key visual elements in remote sensing terminology.

Feature 3: Referring Expression Grounding (Object Localization)

- **Capability:** Locates specific objects requested via text (e.g., 'The dark green solar panel array near the road') and predicts spatial coordinates.
- **Endpoints:** POST /api/refer, POST /api/refer/upload
- **Output:** Normalized bounding box [x1, y1, x2, y2] scaled from [0.0, 1.0].

Feature 4: Multi-Channel Realtime Engine (Firebase + REST)

- **Capability:** Integrated dual-pipeline supporting direct REST HTTP requests and asynchronous web application sync via Firebase Realtime Database and Firebase Storage.
- **Firebase Nodes:** /queries/{id} (incoming queue) & /results/{id} (processed output stream).

4. Technical System Architecture

Layer	Technology Stack	Role & Key Configuration
AI / LLM Model	Qwen2.5-VL-3B + LoRA PyTorch 2.14 · PEFT 0.20	Fine-tuned vision-language model with 4-bit NF4 quantization for low-memory execution.
Backend Framework	FastAPI 0.115+ Uvicorn ASGI Server	Asynchronous REST API, CORS middleware, lifespan events, and openapi documentation.
Realtime DB & Storage	Firebase Admin SDK Realtime DB + Cloud Storage	Background listener service for async client processing and cloud image uploads.
Image Processing	Pillow 12.3 qwen-vl-utils 0.0.14	Image format decoding, base64 extraction, aspect-ratio patch division for Qwen vision encoder.

5. REST API & Endpoint Contracts

Sample Health Check Payload (GET /api/health):

```
{ "status": "ok", "model_loaded": true, "model_name": "Qwen/Qwen2.5-VL-3B-Instruct",  
  "firebase_listener": true, "queries_processed": 142, "errors": 0 }
```

Sample VQA Request & Response (POST /api/vqa):

```
// Request { "image_base64": "data:image/jpeg;base64,...", "question": "What is the primary land  
use in this region?" } // Response { "answer": "The image shows predominantly agricultural  
farmland with circular center-pivot irrigation fields.", "confidence": 0.94, "task": "vqa",  
  "inference_time_ms": 1420 }
```

6. Non-Functional & Security Requirements

- **Data Privacy & Security:** All core inference operations execute locally on-device. No telemetry or imagery data is sent to 3rd-party LLM providers.
- **Scalability:** Stateless FastAPI handlers allow multi-worker horizontal scaling behind load balancers.
- **Reliability & Fallbacks:** Includes stub fallback execution mode (`SATQUERY_MODEL_MODE=stub`) for cloud UI testing when GPU hardware is offline.
- **Maximum File Constraints:** Enforces a hard limit of 20 MB per uploaded satellite image to prevent buffer exhaustion.

7. Product Roadmap & Future Enhancements

- **Phase 1 (Current):** Local Qwen2.5-VL-3B + LoRA deployment, REST API, Firebase Realtime Sync.
- **Phase 2 (Q4 2026):** Multi-temporal change detection (comparing satellite passes across different dates).
- **Phase 3 (Q1 2027):** GeoTIFF metadata parsing (extracting exact lat/long bounding boxes and spatial resolution metadata).
- **Phase 4 (Q2 2027):** Interactive bounding box rendering directly in Web UI canvas.